Final Project Write-Up

GEOG 3023

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4/22/26

Income, Space, and Immigration: Explaining the Distribution of Foreign-Born Populations Across U.S. States

Project Overview

This project examines the relationship between median household income and the percent foreign-born population across the United States. The main analysis uses 2020-2024 American Community Survey data to test whether higher-income states tend to have larger foreign-born populations, using maps, correlation, linear regression, model diagnostics, and spatial autocorrelation tests. I also expanded the project beyond the original scope because this is my last college project, and I got interested in seeing how far I could take the analysis by connecting ideas from different classes. To build on the state-level results, I added a random forest prediction model, Department of Homeland Security lawful permanent resident flow data, and a county-level ACS extension. Overall, the results show that income is related to immigrant settlement patterns, but immigration geography is also shaped by existing immigrant communities, spatial clustering, metropolitan concentration, and different immigration pathways.

Section Overview

This report begins with the required final project questions, which test the relationship between median household income and percent foreign-born population across U.S. states. After the required analysis, I include three additional sections. The first uses a random forest model to explore prediction. The second adds Department of Homeland Security lawful permanent resident data to compare existing immigrant communities with new immigration flows. The third shifts to county-level ACS data to show how immigration patterns become more detailed at a local scale, with a final focus on Colorado counties.

Section 1. Income and Foreign-Born Population Across U.S. States

1. What topic are you studying and why did you select this topic?

I selected immigration and median household income as my final project topic. My main research question is: How does median household income influence the spatial distribution of immigrants across U.S. states?

I chose this topic because I am interested in immigration patterns in the United States and wanted to study whether immigrants are more concentrated in states with stronger economic conditions. I originally considered studying emigration from the United States, but state-level immigration data were more available, easier to connect to spatial data, and more useful for the statistical questions in this project.

2. What are your null and alternative hypotheses?

Null hypothesis: There is no relationship between median household income and the percentage of foreign-born population across U.S. states.

Alternative hypothesis: States with higher median household income have a higher percentage of foreign-born population.

3. What is the dependent/response variable? What is the data source? Provide a map of your response variable.

The dependent or response variable is the percent foreign-born population by state. I calculated this variable as the foreign-born population divided by the total population, multiplied by 100.

The data source is the U.S. Census Bureau American Community Survey 2020-2024 5-year estimates, accessed in R with the tidycensus package. The total population variable is B01003_001 and the total foreign-born population variable is B05006_001. The state geometry came from Census TIGER/Line data through tidycensus and tigris.

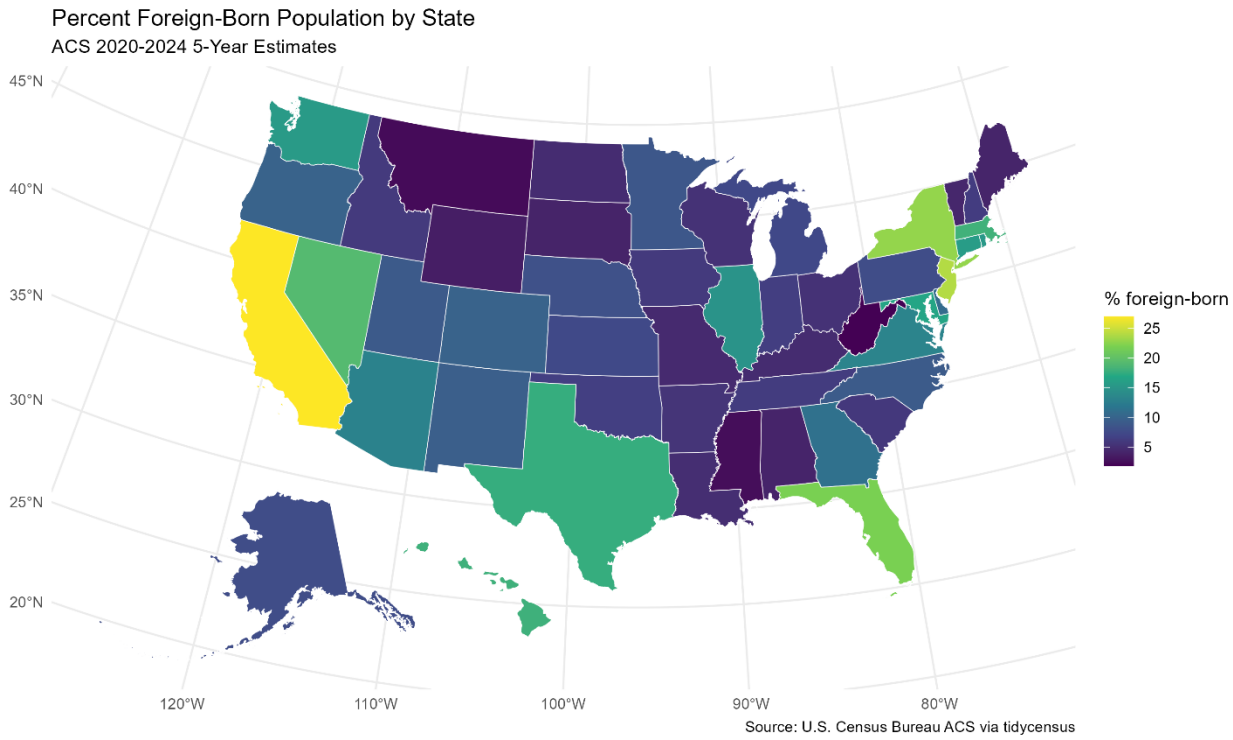


Figure 1. Percent foreign-born population by state

Data source links: HERE and HERE

4. What is your independent/predictor variable? What is the data source? Provide a map of your predictor variable.

The main independent or predictor variable is median household income by state. I used this variable to test whether states with higher income levels also tend to have higher percentages of foreign-born residents.

The data source is also the U.S. Census Bureau American Community Survey 2020-2024 5-year estimates, accessed through tidycensus. The ACS variable for median household income is B19013_001.

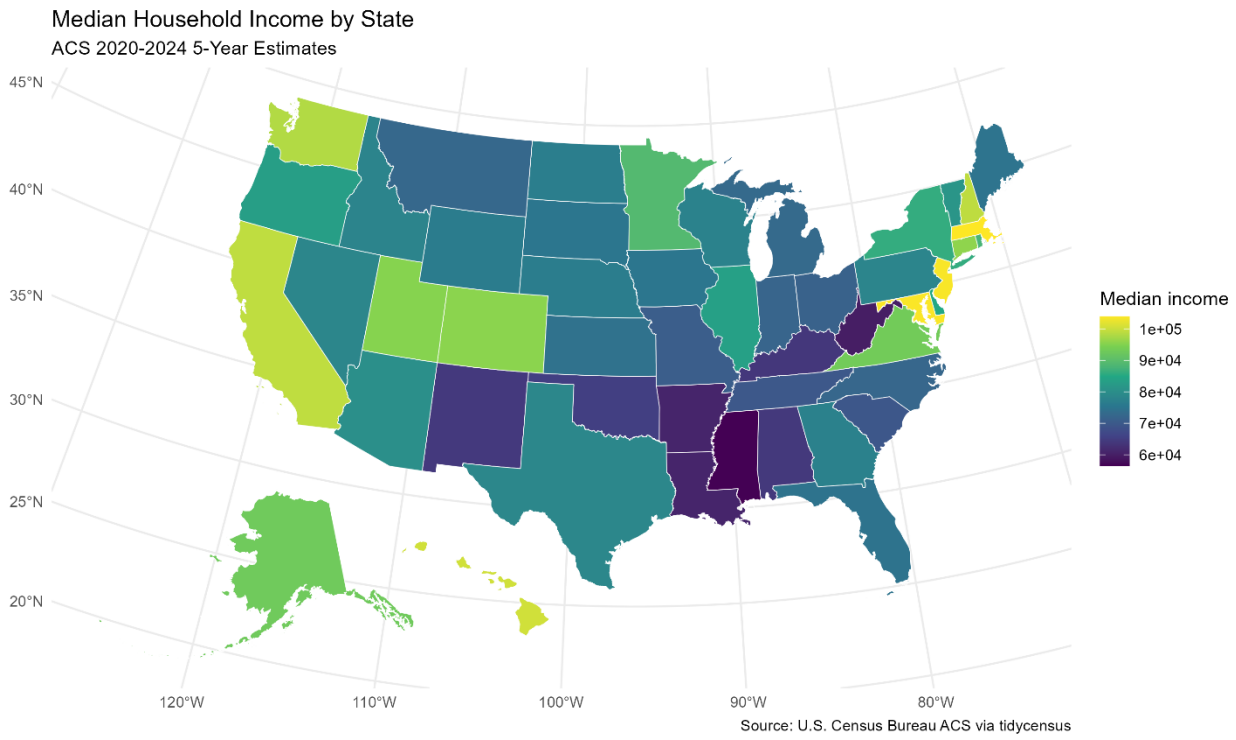


Figure 2. Median household income by state

Data source links: HERE and HERE

5. What did you do to the data to create your dependent and independent variables?

I pulled state-level ACS data directly into R using get_acs() with geometry = TRUE. I removed Puerto Rico and the District of Columbia, so the analysis focused on the 50 U.S. states. The data did not need to be aggregated because the ACS variables were already available at the state level.

To create the response variable, I divided the ACS estimate for total foreign-born population by total population and multiplied by 100. The predictor variable, median household income, was already reported as a state-level ACS estimate, so I kept it as dollars. I also created additional variables for the multiple regression model, including unemployment rate, labor force participation, log population density, and a Mexico border-state dummy variable for California, Arizona, New Mexico, and Texas.

For spatial analysis, I used the state polygon geometry returned by tidycensus/tigris and created queen-contiguity neighbors in R with the spdep package. Alaska and Hawaii do not have contiguous state neighbors, so the spatial weights used zero.policy = TRUE.

6. What are the distributions of your dependent and independent variables?

The average percent foreign-born population across the 50 states was 9.83%, with a standard deviation of 6.26 percentage points. The histogram shows that many states have relatively low to moderate foreign-born percentages, while a smaller number of states have much higher immigrant shares.

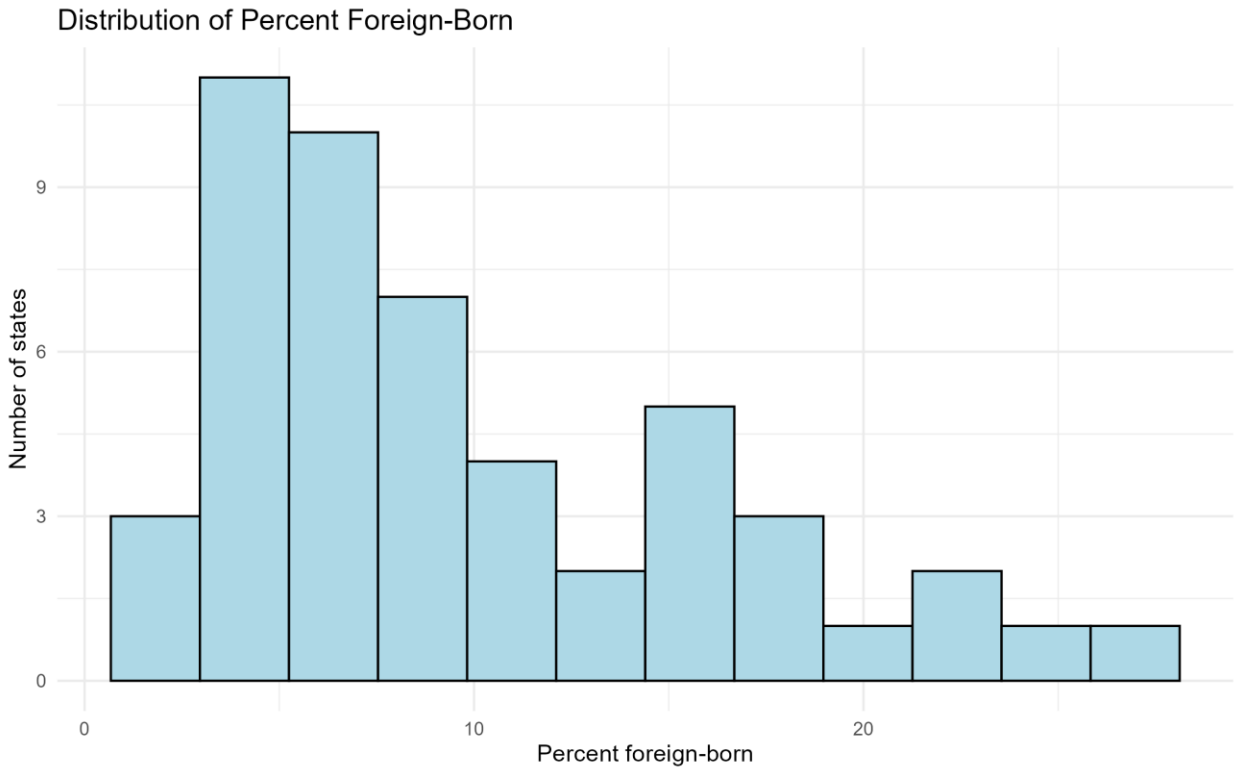


Figure 3. Distribution of percent foreign-born population.

The average median household income was $79,822.42, with a standard deviation of $12,751.49. The income histogram shows variation across states, with most states clustered around the middle of the distribution and fewer states at the very high end.

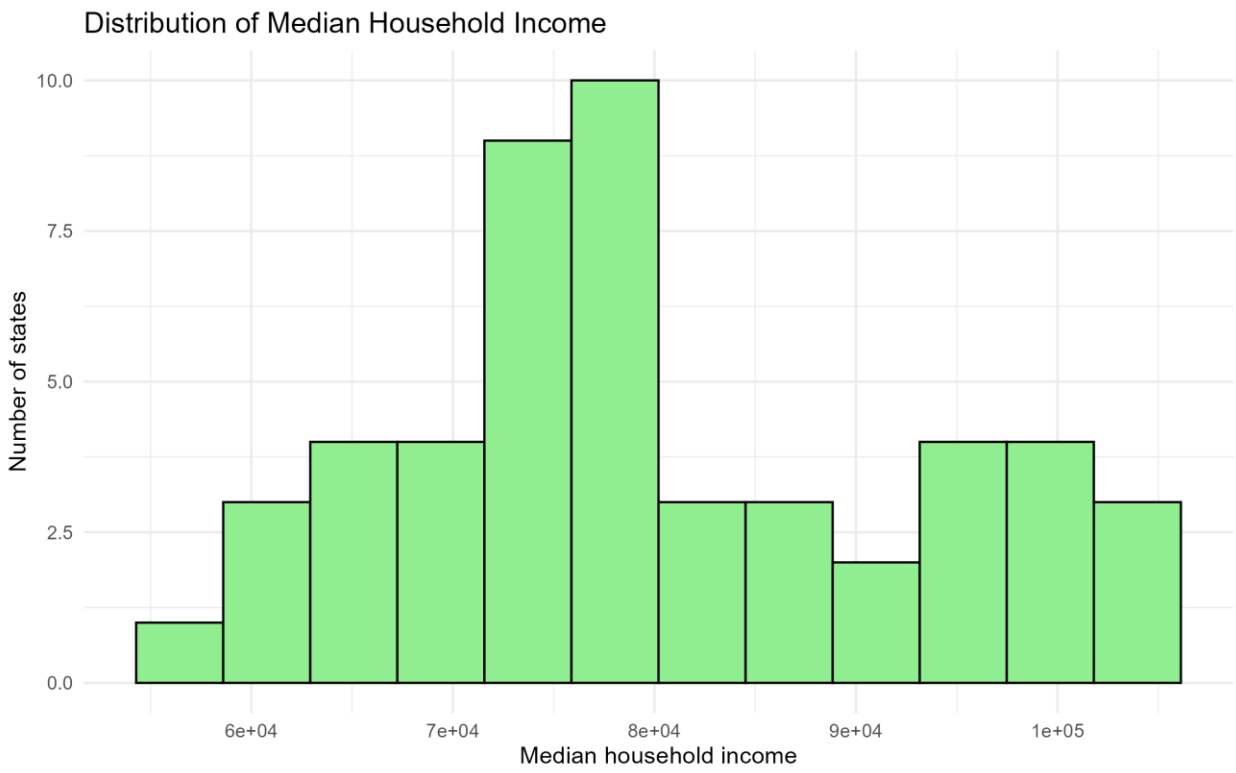


Figure 4. Distribution of median household income

7. Is the relationship between the two variables linear?

The scatter plot shows a positive relationship between median household income and percent foreign-born population. The pattern is not perfect, but the points generally rise from left to right, which suggests that states with higher median income tend to have higher immigrant percentages.

The Pearson correlation coefficient was r = 0.648, with a p-value of 3.545e-07. This is a moderately strong positive correlation and it is statistically significant.

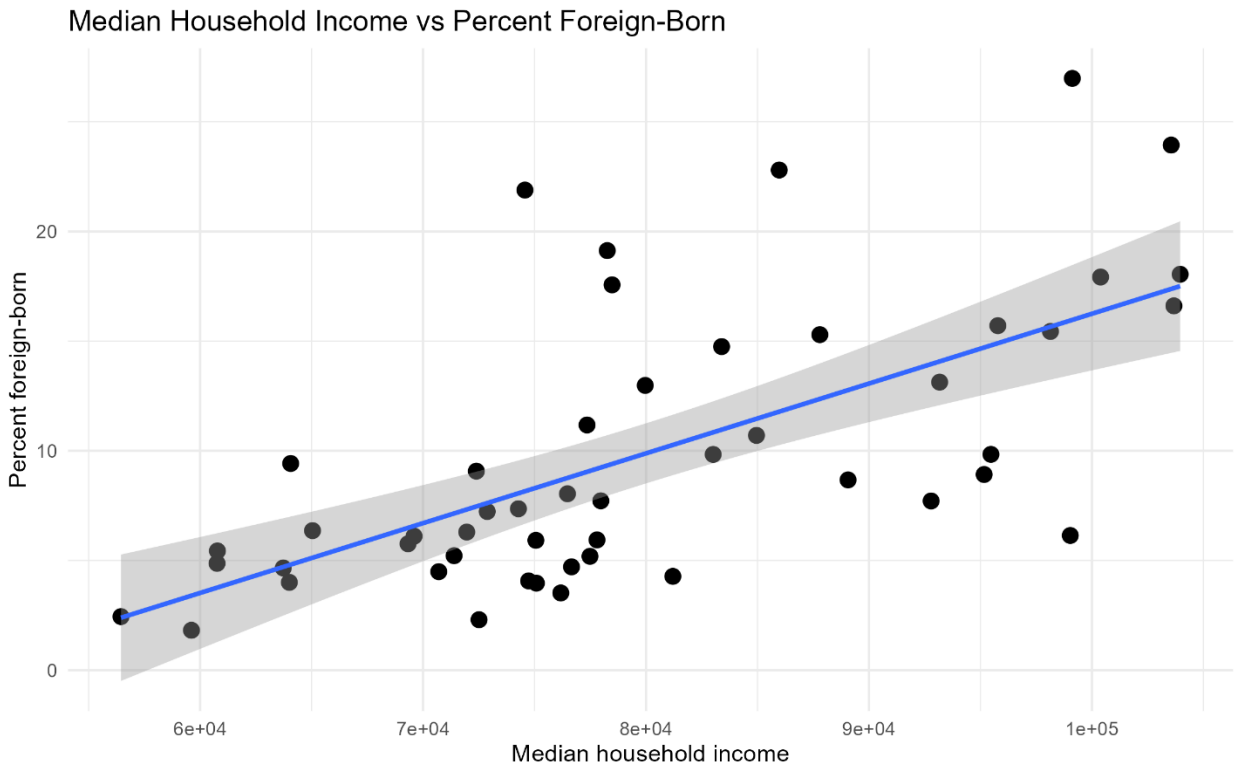


Figure 5. Relationship between median household income and percent foreign-born population.

8. Is your predictor variable a strong predictor of your response variable?

In the simple linear regression, median household income was a statistically significant predictor of percent foreign-born population. The coefficient for median income was 0.0003182, with a p-value of 3.55e-07. This means that, in the simple model, higher median household income is associated with a higher percent foreign-born population.

The simple model R-squared was 0.4205, and the adjusted R-squared was 0.4084. This means median household income alone explained about 42% of the variation in state foreign-born percentages. I would describe this as a meaningful predictor, but not a complete explanation.

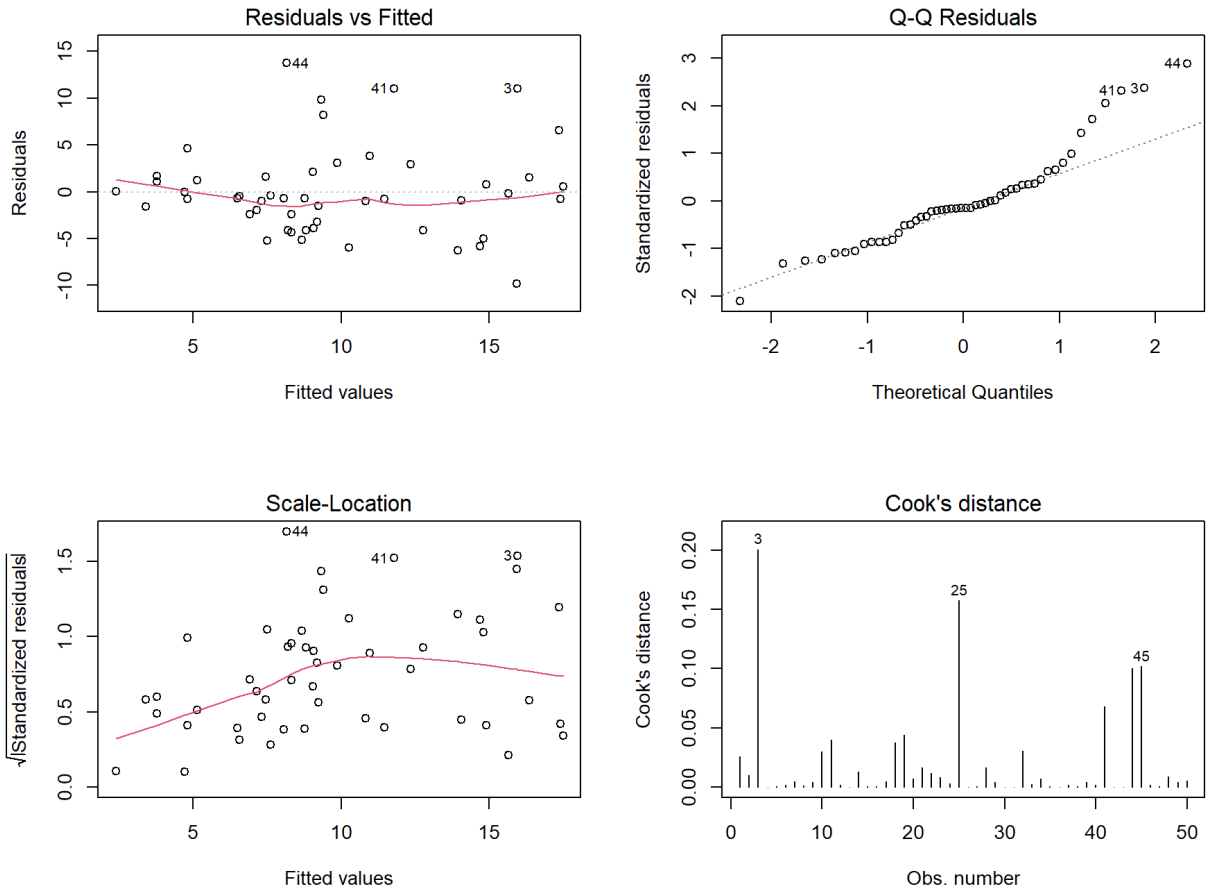


Figure 6. Linear model diagnostic plots for the simple regression model

I also ran a multiple regression model with median income, unemployment rate, labor force participation, log population density, and border-state status. This model had an R-squared of 0.7680 and an adjusted R-squared of 0.7417, so it explained substantially more variation than the simple model.

For the four linear model assumptions, the linearity assumption was mostly supported by the scatter plot, although the relationship is not perfect. The diagnostic plots suggested possible outliers and uneven residual spread. The Shapiro-Wilk tests showed that residual normality was not fully met for either model: the simple model residuals had p = 0.004129 and the multiple model residuals had p = 0.0001928. Independence was also a concern in the simple model because the residuals had significant spatial autocorrelation. These limitations mean the regression results are useful for describing the relationship, but they should not be interpreted as proving a causal relationship.

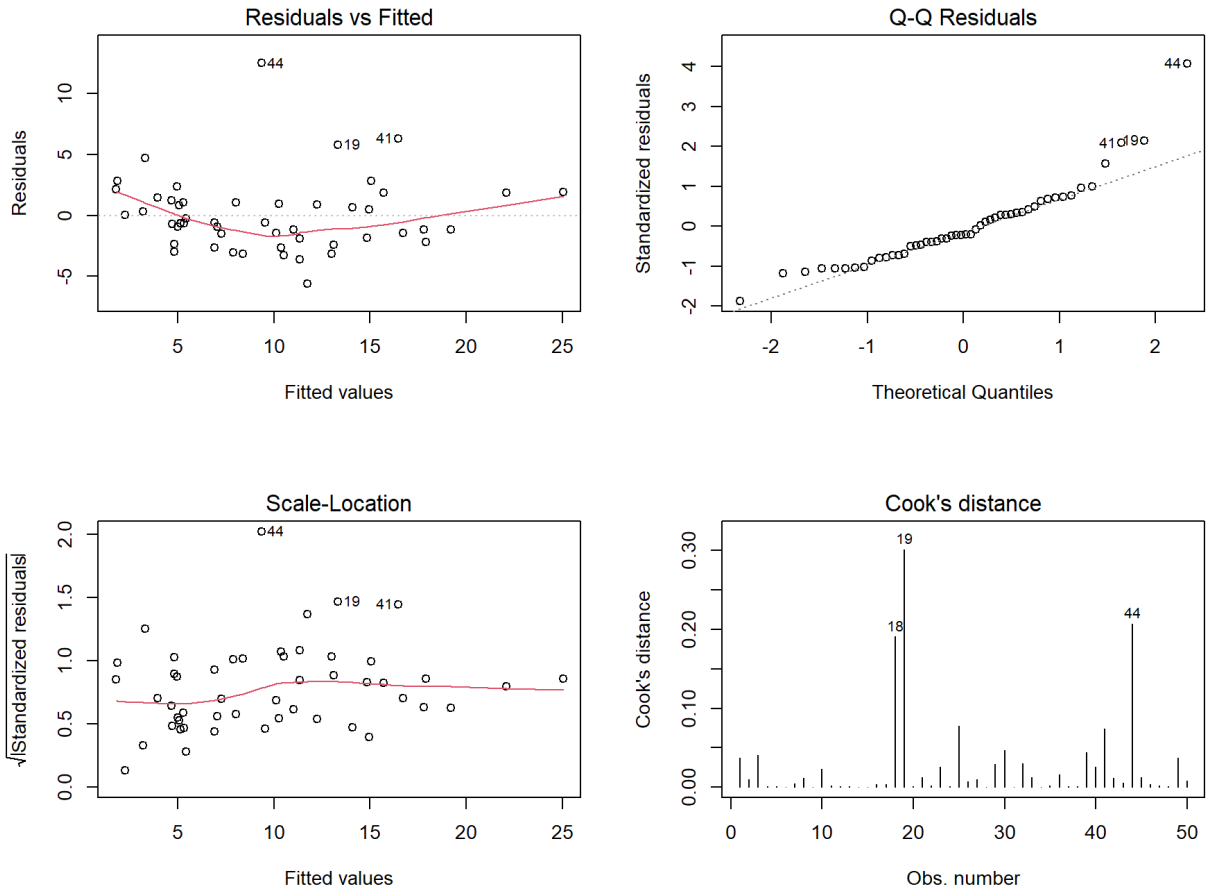


Figure 7. Linear model diagnostic plots for the multiple regression model

9. Do your response variable and predictor variables show spatial dependence or spatial independence?

The response variable showed spatial dependence. Moran's I for percent foreign-born population was 0.2599 with a p-value of 0.001888, meaning nearby states tended to have more similar immigrant percentages than expected under spatial randomness.

The main predictor variable also showed spatial dependence. Moran's I for median household income was 0.3028 with a p-value of 0.0004711. Log population density had even stronger spatial clustering, with Moran's I = 0.5367 and p = 3.489e-09.

The simple model residuals still showed significant spatial autocorrelation, with Moran's I = 0.2578 and p = 0.001831. This means the simple income-only model missed some spatial

pattern. The multiple model residuals were much less spatially autocorrelated, with Moran's I = 0.0659 and p = 0.1713, so adding unemployment, labor force participation, population density, and border-state status improved the model.

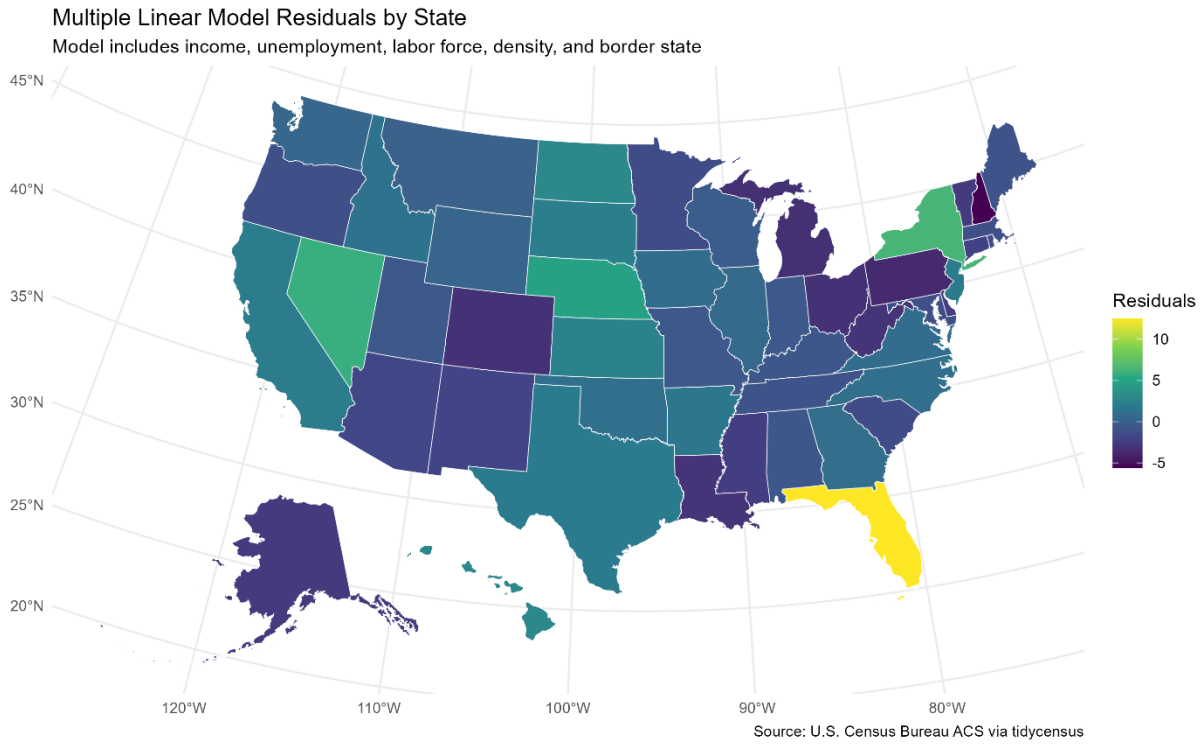


Figure 8. Simple regression residuals by state

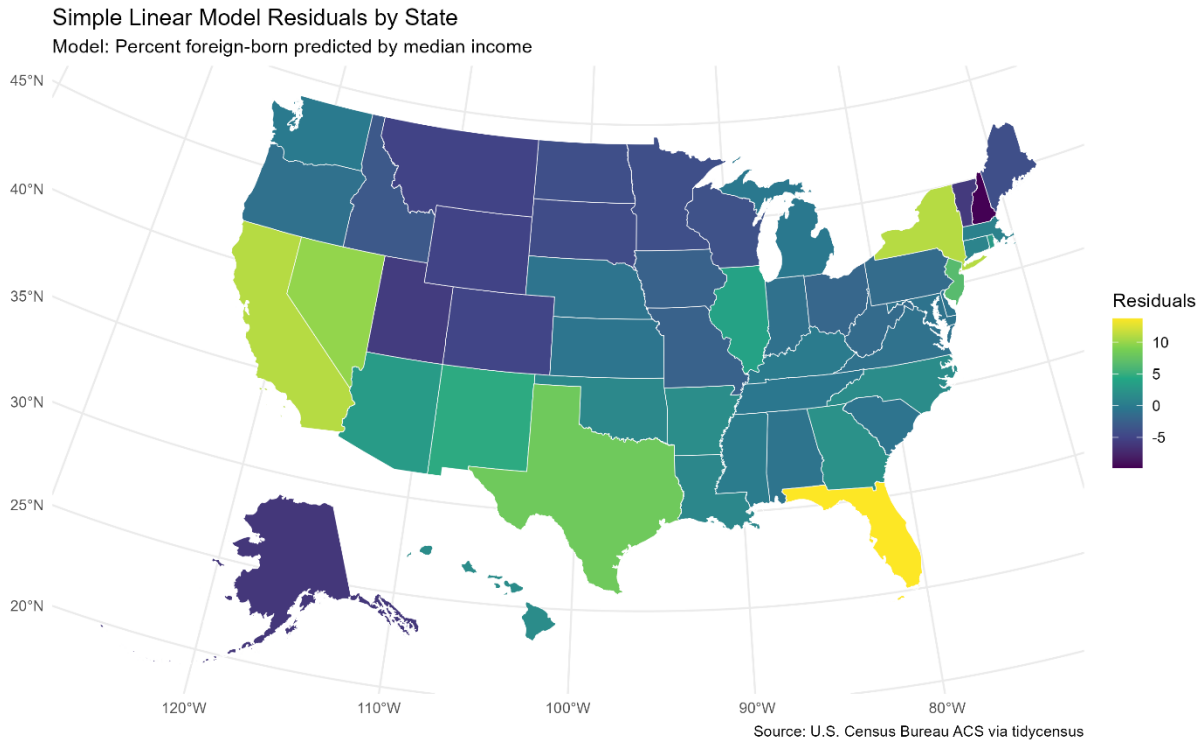


Figure 9. Multiple regression residuals by state

If I were going to improve the model further, I would include spatial variables such as region, contiguity-based spatial lag terms, or a spatial regression model. This would help account for the fact that immigration patterns are geographically clustered instead of randomly distributed across states.

10. What are your conclusions about the relationship between your response and predictor variables?

My conclusion is that median household income and percent foreign-born population have a positive and statistically significant relationship across U.S. states. States with higher median household income generally had higher foreign-born population percentages. In simple terms, states with higher household incomes tended to have higher immigrant populations, but income is only one part of the explanation because immigration is also affected by jobs, location, housing, family networks, and other social factors. The Pearson correlation was r = 0.648, and the simple regression p-value for median income was 3.55e-07.

Because the relationship was statistically significant, I reject the null hypothesis that there is no relationship between median household income and percent foreign-born population. However, I do not conclude that higher income directly causes immigration. Immigration patterns are also shaped by job markets, housing costs, existing immigrant communities, border location, urbanization, state history, and federal immigration policy. The multiple model showed that a broader set of variables explained the pattern better than income alone.

This matters because immigration is a spatial and social process, not just an income process. The final results support the idea that economic opportunity is related to where immigrants live, but they also show that spatial context needs to be included for a stronger explanation.

Section 2: Additional Analysis 1 - Random Forest Prediction Model

As a first additional analysis, I tested a random forest model to see whether a larger set of state-level variables could predict percent foreign-born population better than the original linear regression model. The original model focused mainly on median household income, but immigration patterns are likely shaped by several overlapping factors. For this model, I included median income, unemployment rate, labor force participation, log population density, border-state status, coastal-state status, bachelor's degree percentage, median rent, and language other than English spoken at home.

The random forest model performed well as a prediction model. It explained 88.22% of the variance in the model output, with an RMSE of 0.944, MAE of 0.668, and prediction R-squared of 0.986. This means the model's predicted foreign-born percentages were very close to the actual values for most states. However, this does not mean the model proves what causes immigration. Random forest models are useful for prediction, but they are harder to interpret than linear regression models.

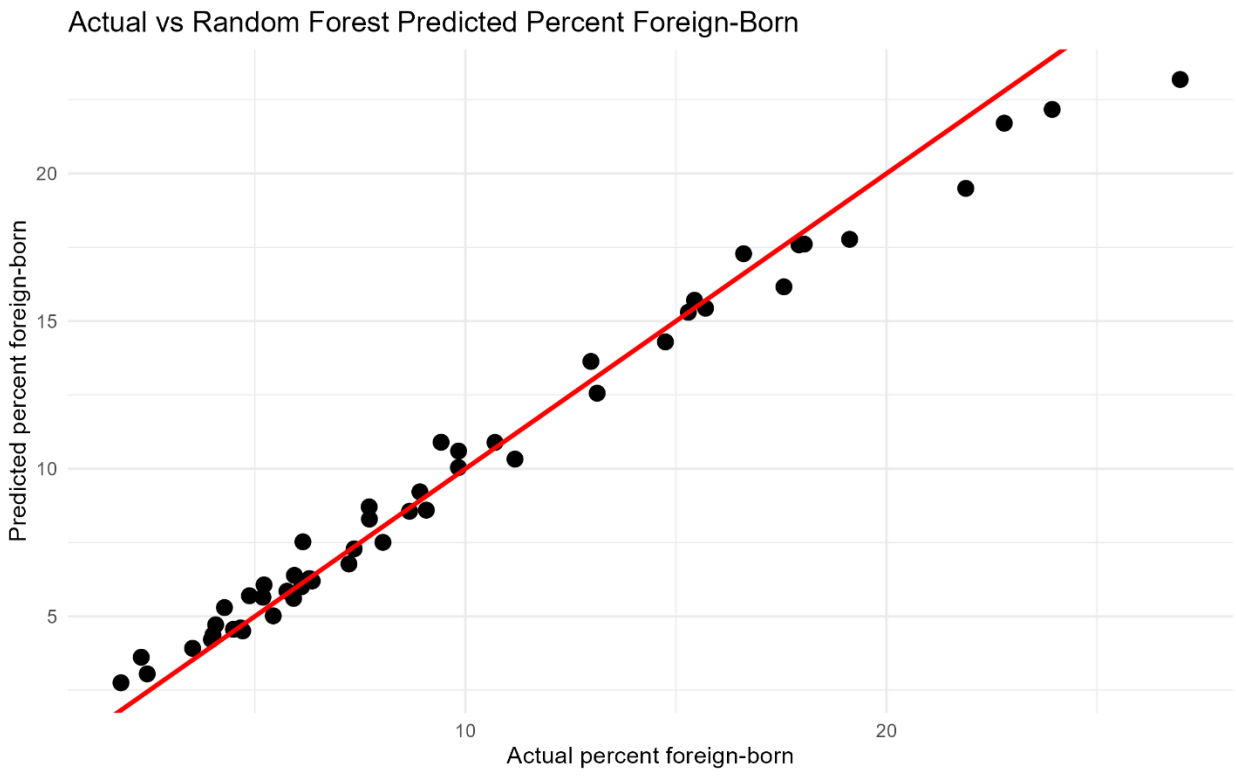


Figure 10. Actual vs. predicted percent foreign-born population from the random forest model.

The most important predictor was language other than English spoken at home, followed by median rent, unemployment rate, log population density, and median income. This is interesting because it suggests that existing immigrant communities and language networks may be very important for predicting where immigrants live. At the same time, this result should be interpreted cautiously because language spoken at home is closely related to immigration itself, so it may partly capture the outcome I am trying to explain.

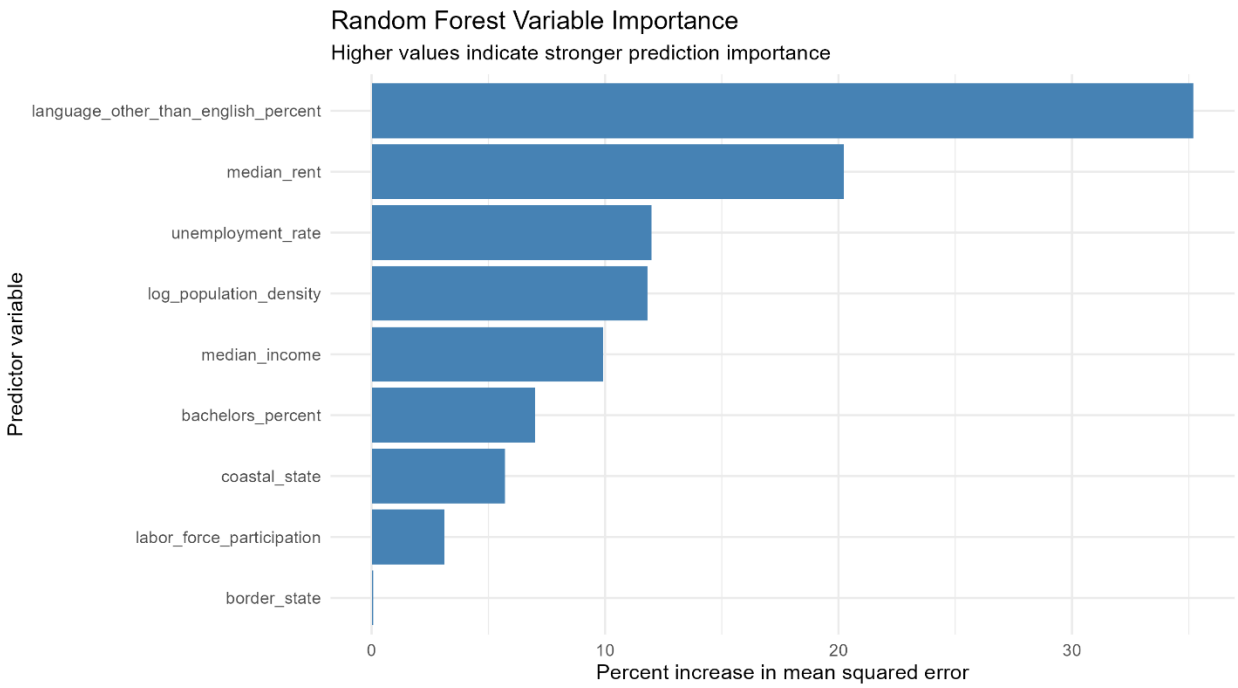


Figure 11. Random forest variable importance for predicting foreign-born population

Overall, the machine learning model helped show that immigration patterns are not explained by income alone. It predicted state foreign-born percentages well, but it was better as an exploratory tool than as a causal explanation. Because the dataset only includes 50 states, I treated this model as a way to identify possible patterns rather than as final proof of what drives immigration. The prediction error map shows where the random forest overpredicted or underpredicted foreign-born population share, which helps identify states where the model fit was weaker.

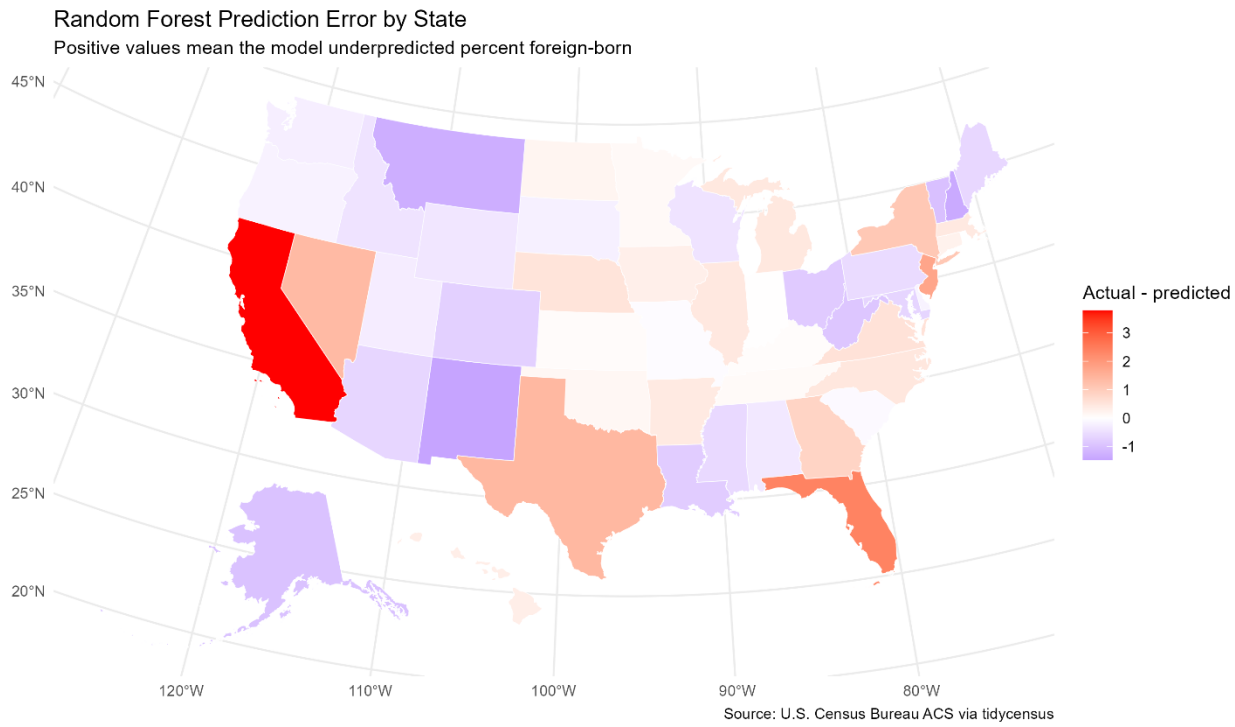


Figure 12. Random forest prediction error by state

Section 3: Additional Analysis 2 - Adding DHS Immigration Flow Data

My project changed as I worked through the analysis. In the original Final folder, I focused on the basic relationship between median household income and percent foreign-born population across U.S. states. In Final 2, I improved the project by adding better ACS variables, maps, residual analysis, spatial autocorrelation tests, and birth-region variables. After that, I added a machine learning section using a random forest model. The machine learning model was useful because it showed that a larger set of variables could predict foreign-born population share well, but it was still mainly a predictive model rather than a clear explanation of why immigration patterns happen.

Because of that limitation, I added Department of Homeland Security data on new lawful permanent residents in fiscal year 2023 as a second additional analysis. This improved the project because the Census ACS data measures the existing foreign-born population living in each state, while DHS data measures new lawful permanent residents entering or adjusting status in a specific year. These are not the same measure, but comparing them helps show whether new immigration flows are reinforcing existing immigrant settlement patterns. In other words, the DHS section gives the project a better way to connect current immigrant communities with new immigration flows.

Data source link: HERE

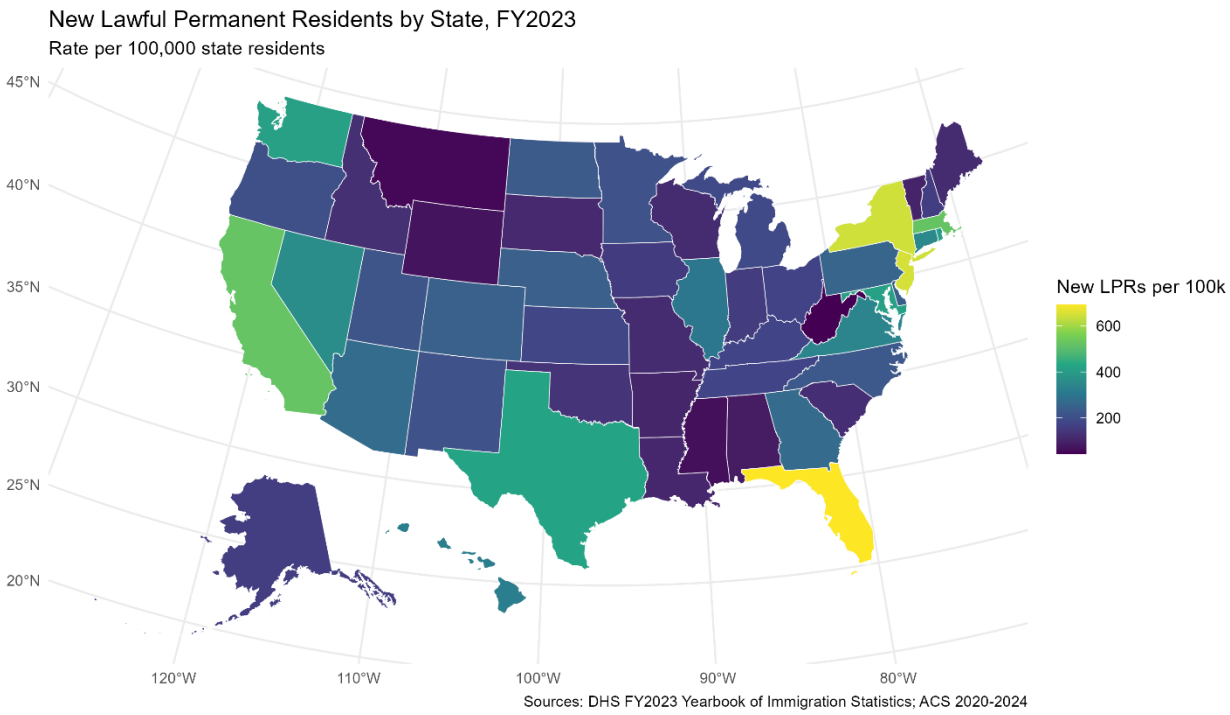


Figure 13. New lawful permanent residents by state, FY2023

The results showed a very strong relationship between a state's existing foreign-born population share and its new LPR rate in 2023. The Pearson correlation was r = 0.946, and the regression model had an R-squared of 0.895. In simple terms, states that already had larger foreign-born populations also tended to receive more new lawful permanent residents per 100,000 residents. This supports the idea that existing immigrant communities may help shape where new immigrants settle.

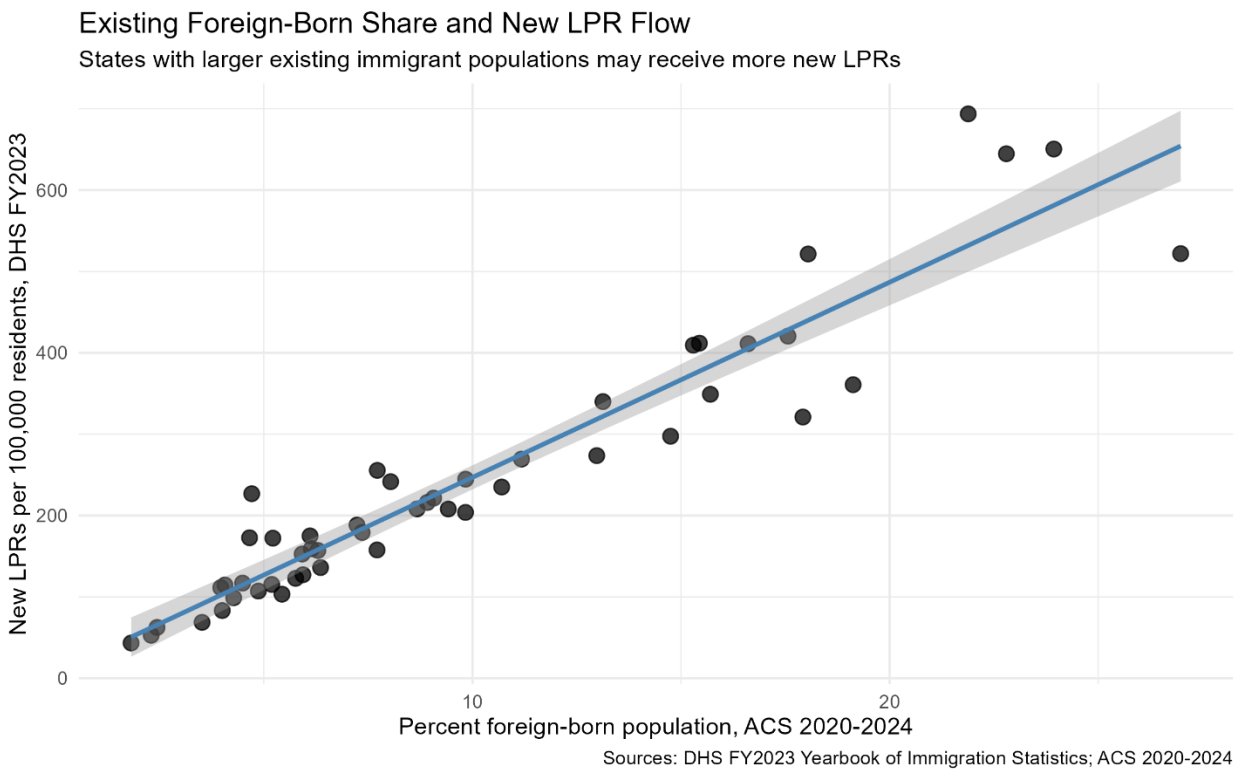


Figure 14. Existing foreign-born share and new lawful permanent resident flow by state

I also compared this relationship by birth region. The relationship was strong for every region I tested. The R-squared values were 0.898 for Africa, 0.871 for Asia, 0.878 for Europe, and 0.858 for Latin America. This means that existing birth-region communities in a state were strongly related to new DHS lawful permanent resident flows from the same region. This adds an important explanation to the original income model: immigration patterns are not only about state income levels, but also about settlement networks and existing communities.

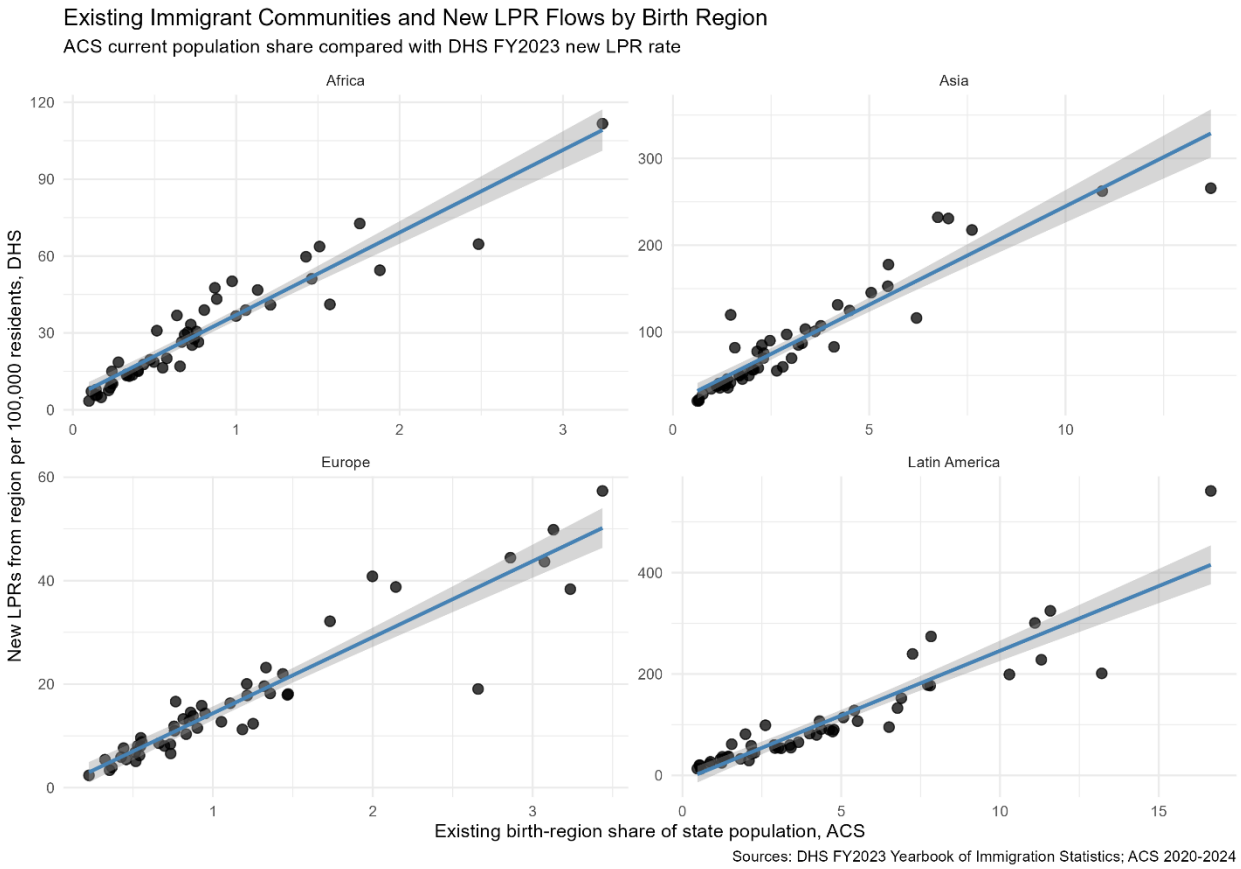


Figure 15. Existing immigrant communities and new lawful permanent resident flows by birth region

This additional analysis strengthens my main conclusion. Median household income is related to foreign-born population patterns, but the DHS results show that existing immigrant communities are also a major part of the story. States with larger immigrant populations are not just higher in foreign-born share; they are also continuing to receive more new lawful permanent residents. This suggests that immigration geography is shaped by both economic opportunity and network effects.

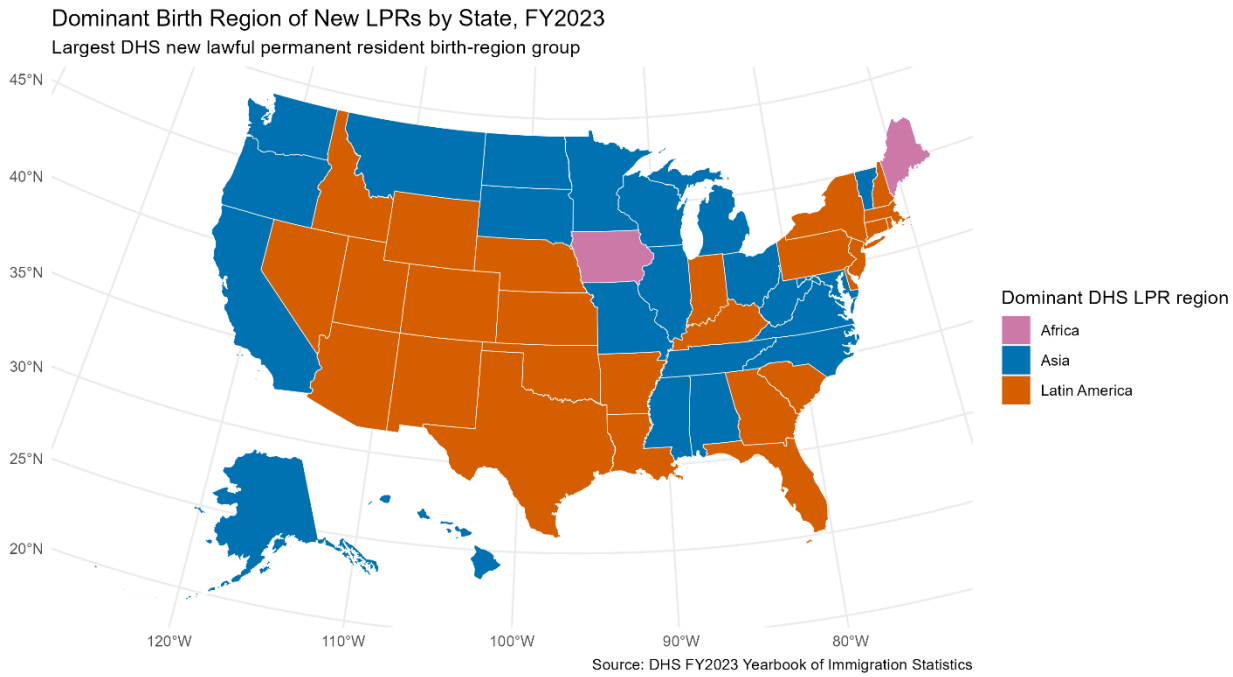


Figure 16. Dominant birth region of new lawful permanent residents by state, FY2023.

I also used DHS admission-class data to better understand how new lawful permanent residents differ by birth region. Latin America had the largest number of new LPRs in 2023, with 550,570 total. For Latin America, the largest category was immediate relatives of U.S. citizens at 54.6%, followed by family-sponsored preferences at 21.5%. Asia and Europe had much higher employment-based shares, at 26.6% and 27.8%, compared with 8.2% for Latin America. Africa had the highest diversity visa share at 23.9% and a higher refugee/asylee share at 14.2%. This shows that different birth regions are connected to different immigration pathways, which helps explain why their state-level patterns are not identical.

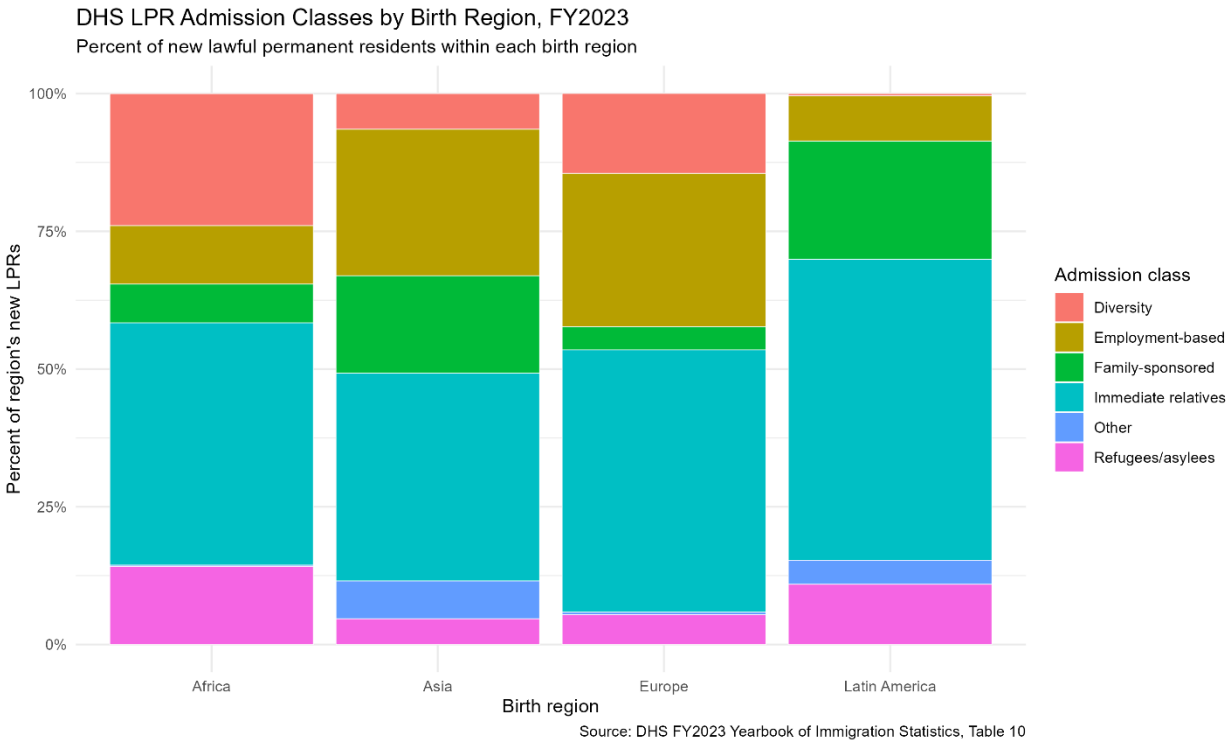


Figure 17. DHS lawful permanent resident admission classes by birth region, FY2023

Section 4: Additional Analysis 3 - County-Level Extension - Local Immigration Patterns Across the United States and Colorado

To strengthen the project, I added a county-level ACS analysis. My original analysis used states, which was useful for comparing broad patterns, but state-level data can hide important local variation. Large states like California, Texas, Florida, and New York contain many different counties with very different immigration patterns. By shifting to counties, I was able to analyze 3,143 counties instead of only 50 states, which gives a more detailed view of where foreign-born populations are concentrated.

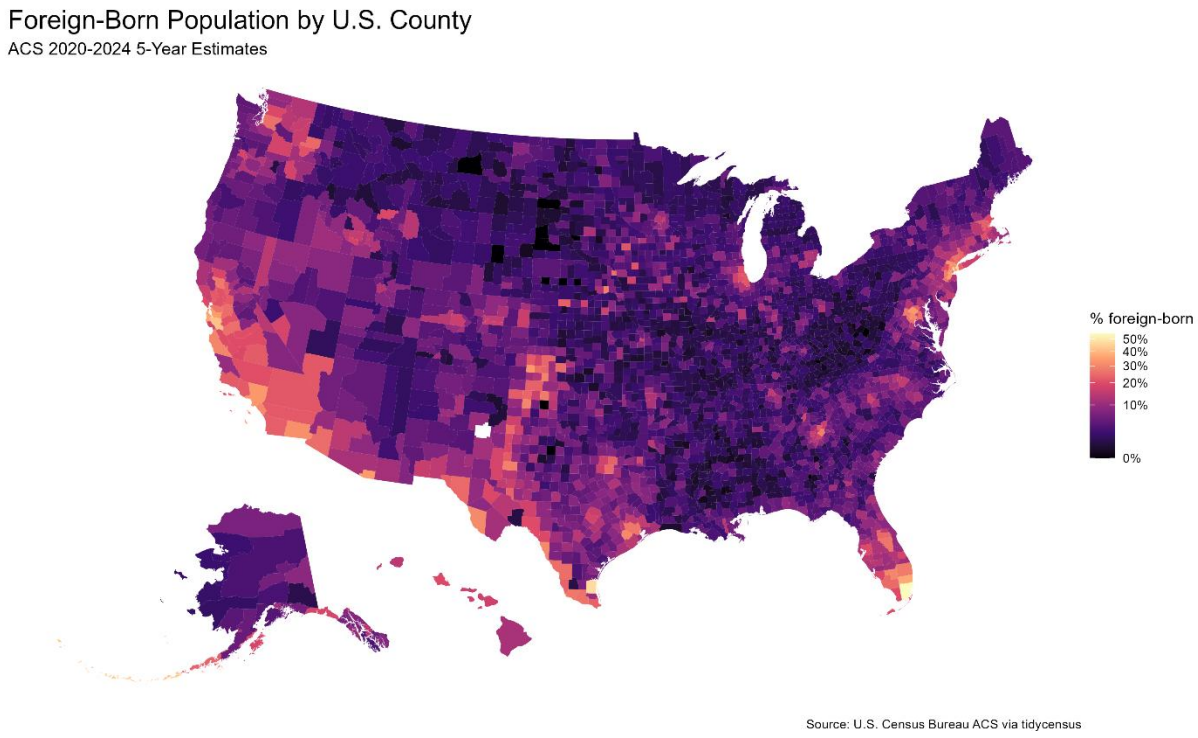


Figure 18. Foreign-born population by U.S. County

The national county map shows that foreign-born populations are not evenly distributed across the United States. Instead, they are concentrated in major metropolitan areas and high-population counties. The top five immigrant destination counties by total foreign-born population were Los Angeles County, California; Miami-Dade County, Florida; Harris County, Texas; Cook County, Illinois; and Queens County, New York. This supports the idea that immigration is strongly connected to large urban labor markets, established immigrant communities, and major metro regions.

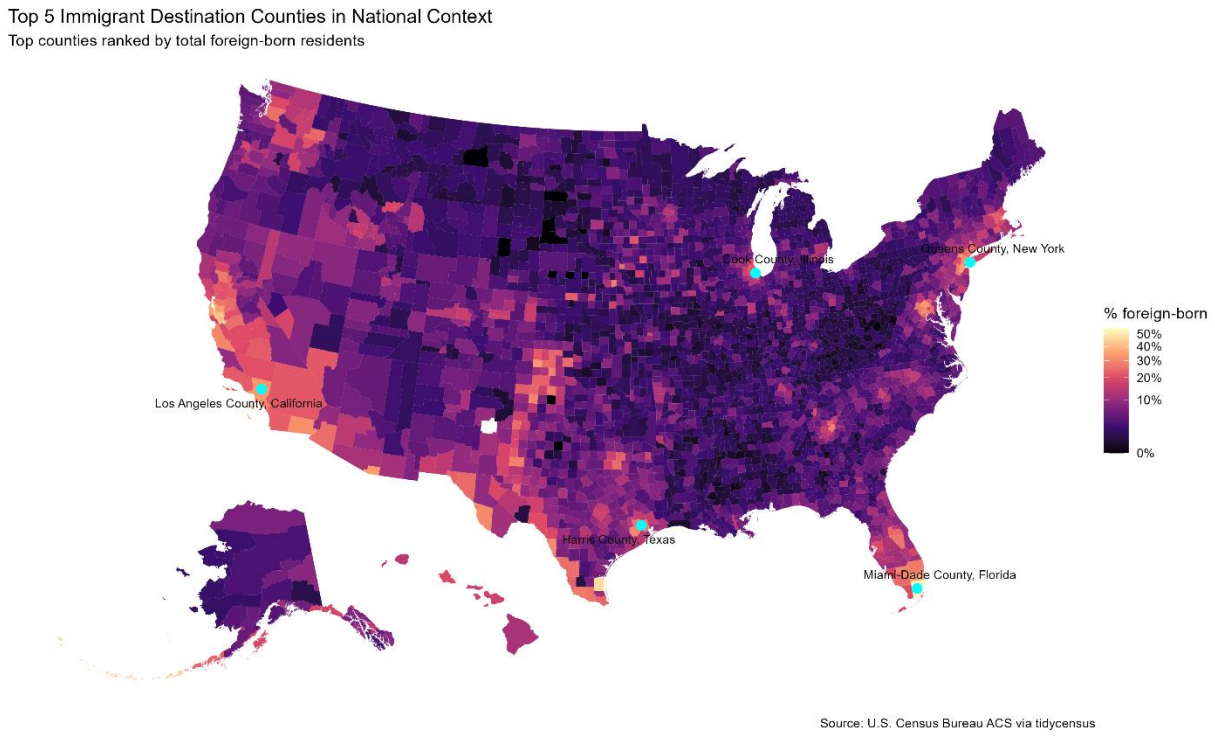


Figure 19. Top 5 immigrant destination counties in national context

I also ran a county-level regression model to test whether county characteristics were related to foreign-born population share. The model included median household income, bachelor's degree percentage, unemployment rate, and log population density. All four predictors were statistically significant, and the model had an R-squared of 0.2478. This means the model explained about 25% of the variation in foreign-born population share across U.S. counties.

This result is important because it shows that the county-level relationship is real, but also more complex than the state-level pattern. Income, education, unemployment, and population density are all related to county-level immigrant settlement patterns, but they do not explain everything. Other factors, such as family networks, housing availability, local labor markets, immigration history, and community support, likely also play a major role.

For the Colorado highlight, Arapahoe County had both the highest foreign-born percentage and the highest total foreign-born population among Colorado counties. The Colorado county map shows that foreign-born populations are more concentrated around the Denver metropolitan area than in rural parts of the state. This makes sense because metro areas usually have more jobs, larger existing immigrant communities, more housing options, and more services.

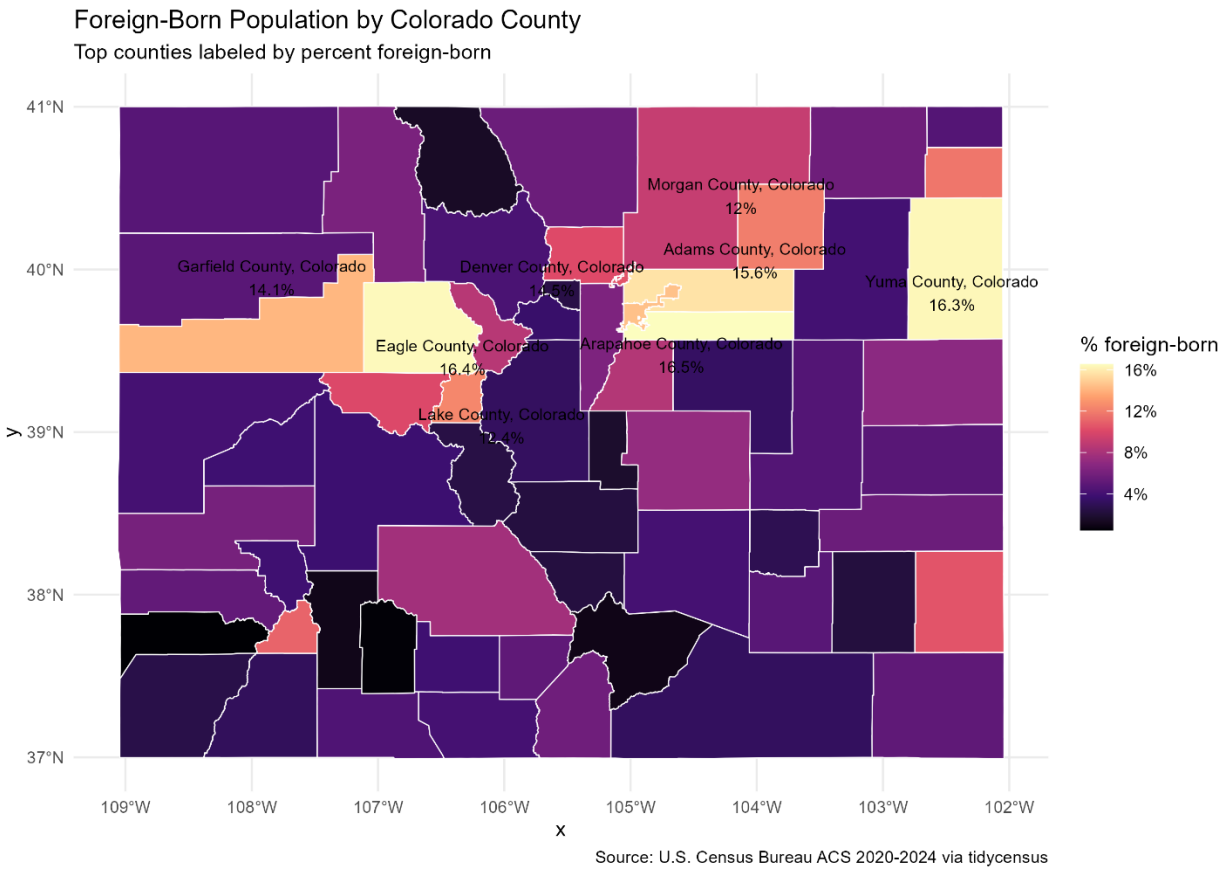


Figure 20. Foreign-born population by Colorado county

The Colorado scatter plot adds a local comparison to the national results. It shows how median household income and foreign-born share vary across Colorado counties. This part of the analysis is useful because it connects the national project back to a local geography. Overall, the county-level extension shows that immigration patterns are not just state-level patterns. They are highly local, and counties reveal important differences that are hidden when the analysis only uses states.

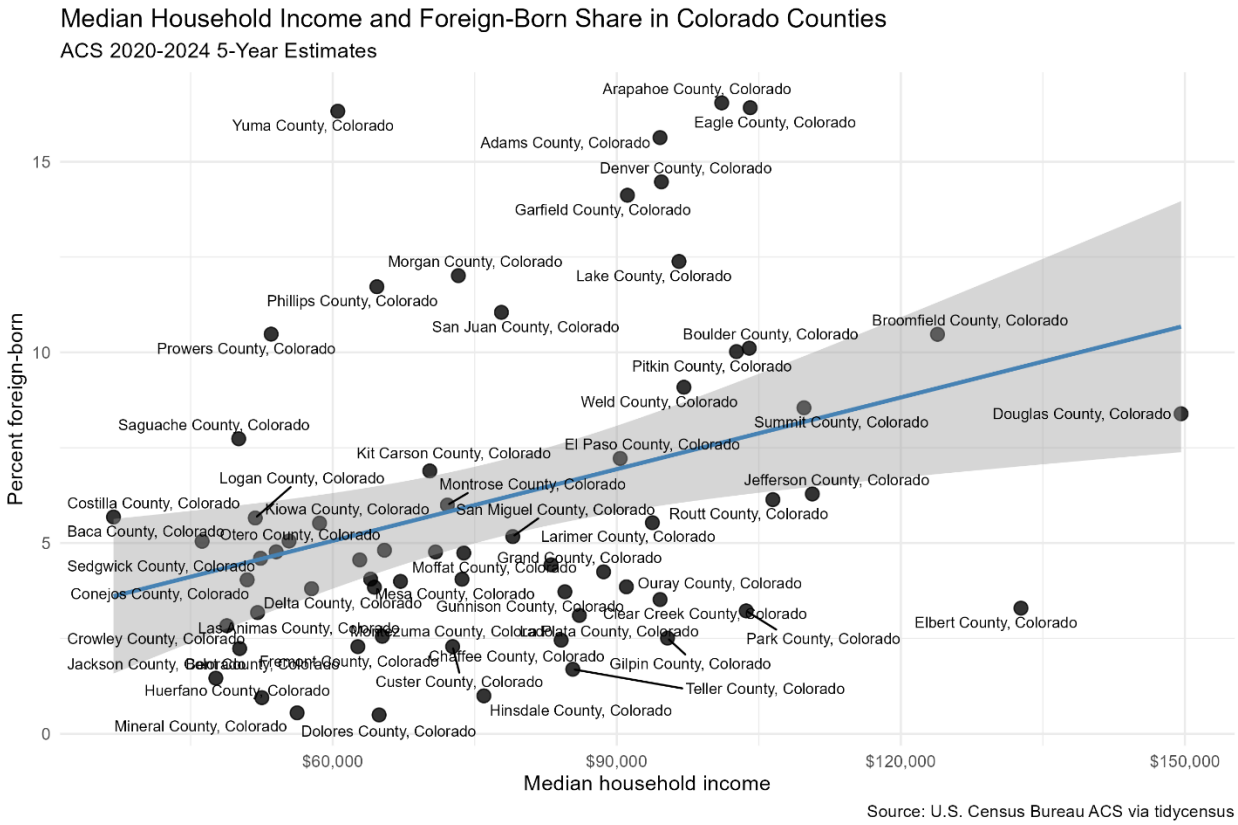


Figure 21. Median household income and foreign-born share in Colorado counties